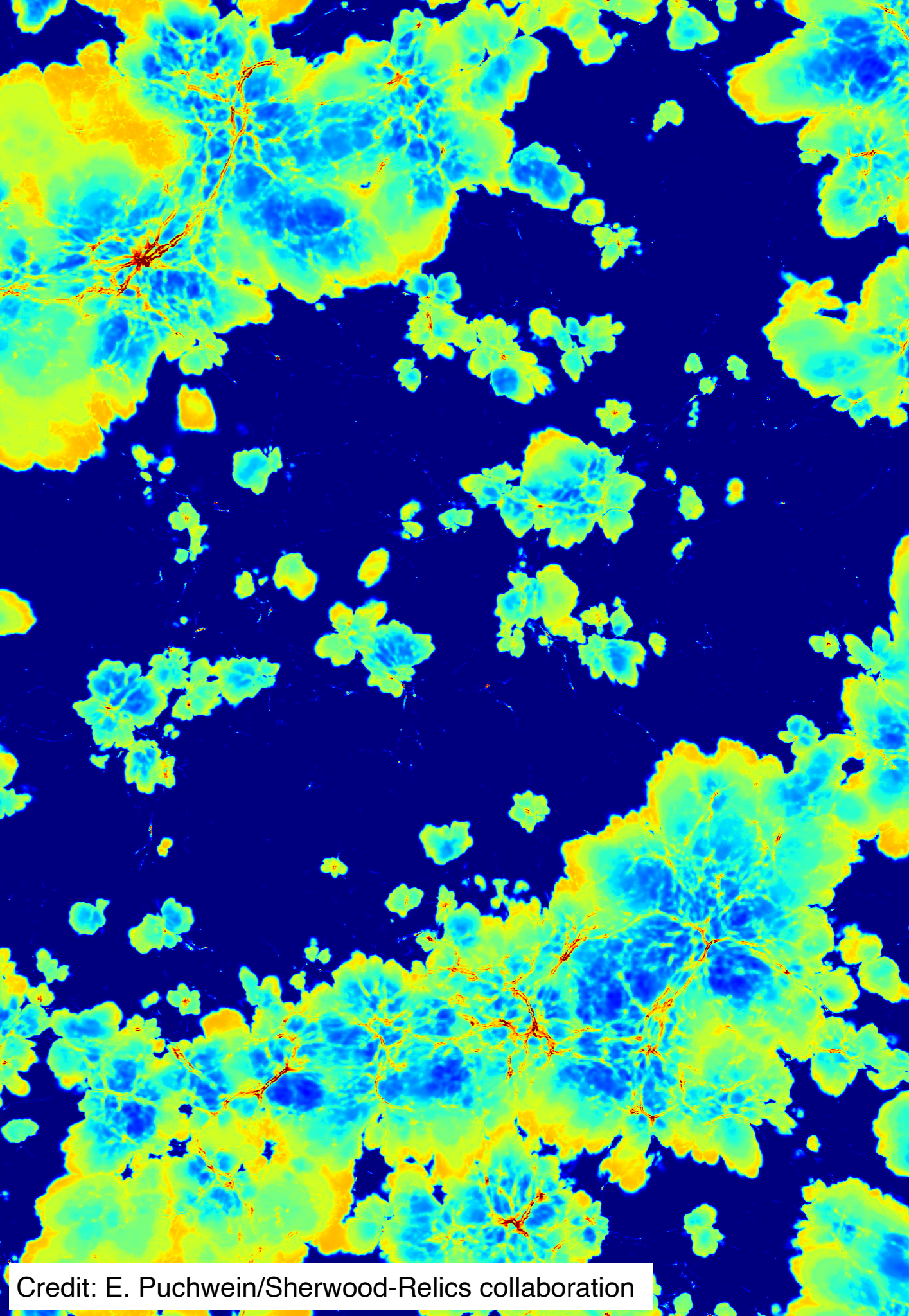


A visualization of the cosmic web, showing a complex network of filaments and clusters of matter. The filaments are colored in bright yellow and orange, while the clusters are in a darker blue. The background is a deep black.

Port Douglas, 24.09.25

Exploring the reionisation era IGM with Ly- α absorption

Jamie Bolton



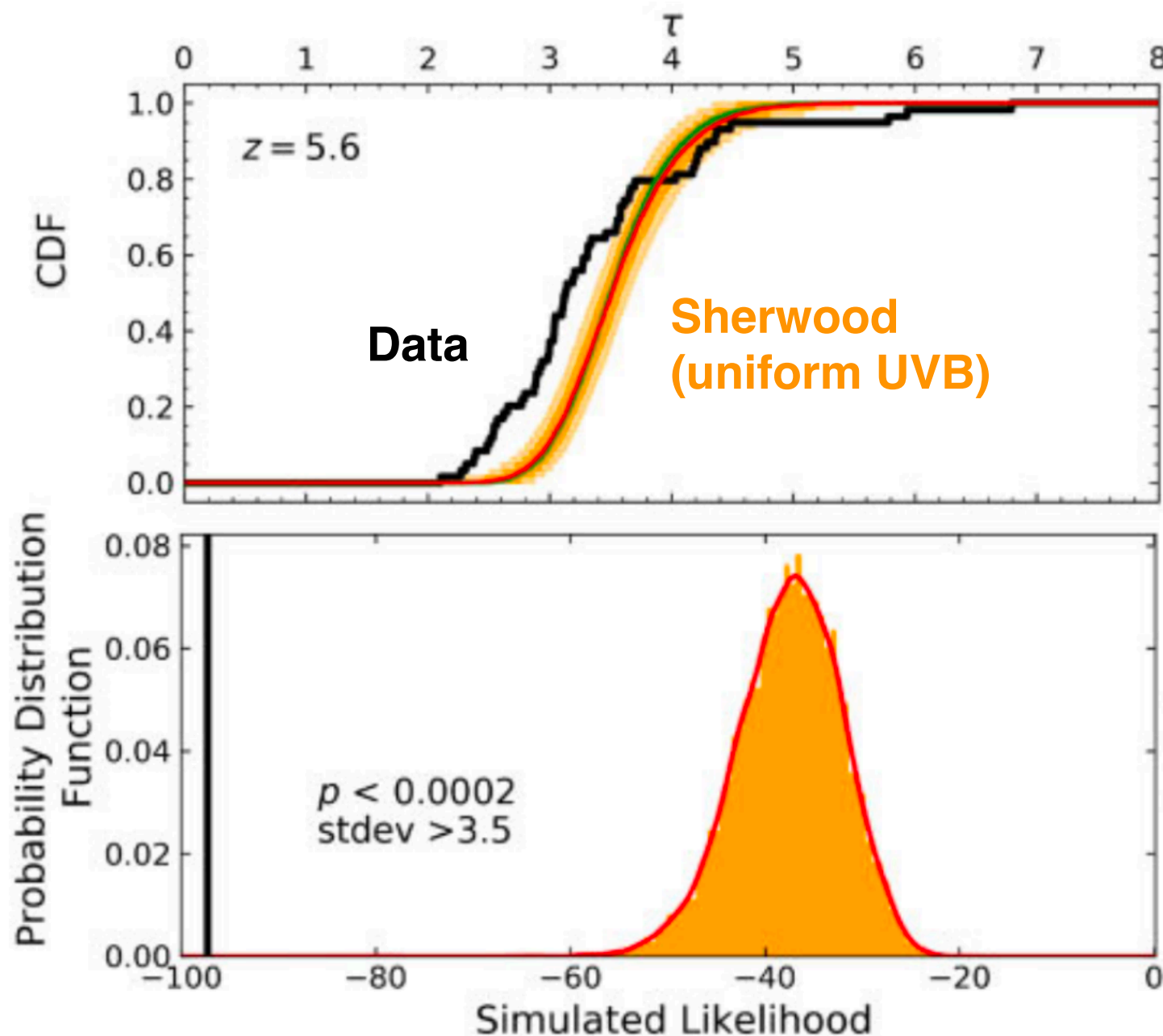
University of
Nottingham
UK | CHINA | MALAYSIA

with **Shikhar Asthana**, George Becker,
Luke Conaboy, **Jen Feron**, Prakash
Gaikwad, Martin Haehnelt, Vid Iršič, Laura
Keating, Girish Kulkarni, **Ke (Mark) Ma**,
Ewald Puchwein, **Fiona Sawyer**

Credit: E. Puchwein/Sherwood-Relics collaboration

- I. Progress
- II. Challenges
- III. Looking ahead

I. Ly- α forest transmission

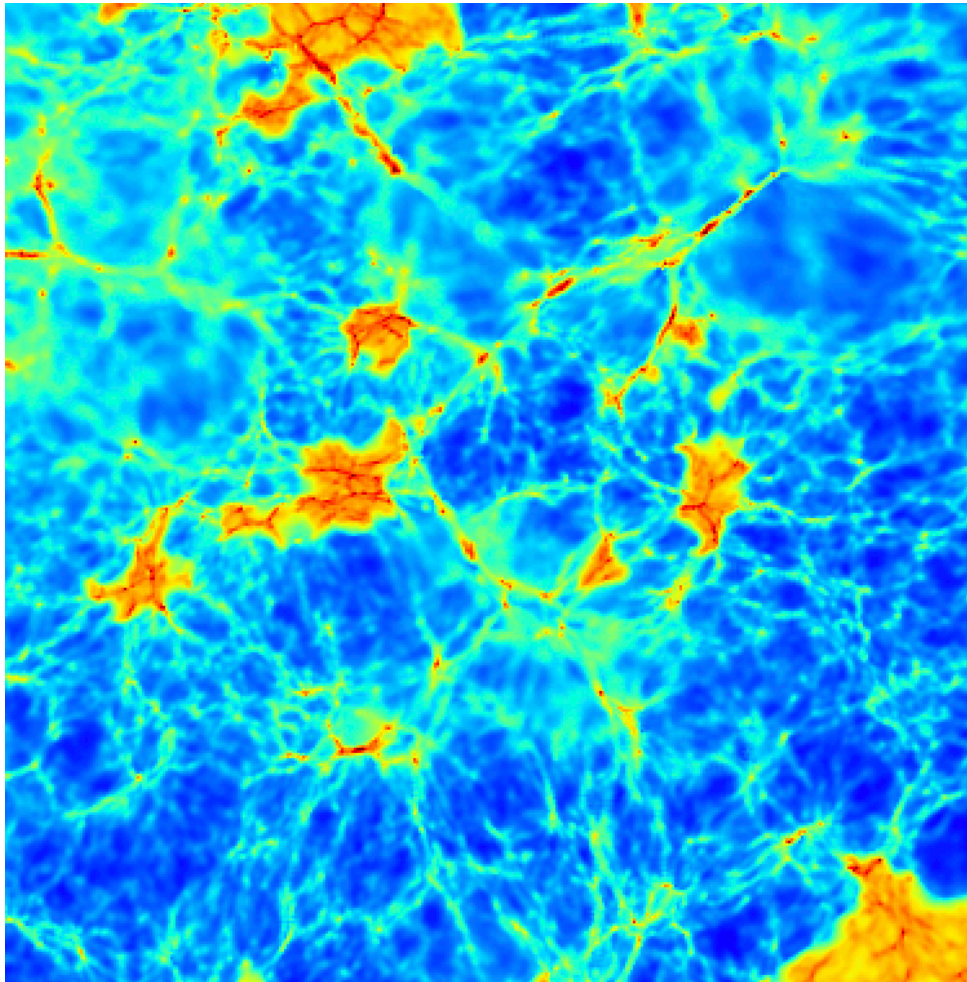


- Distribution of Ly- α forest transmission inconsistent with spatially uniform UV background at $z \geq 5.3$

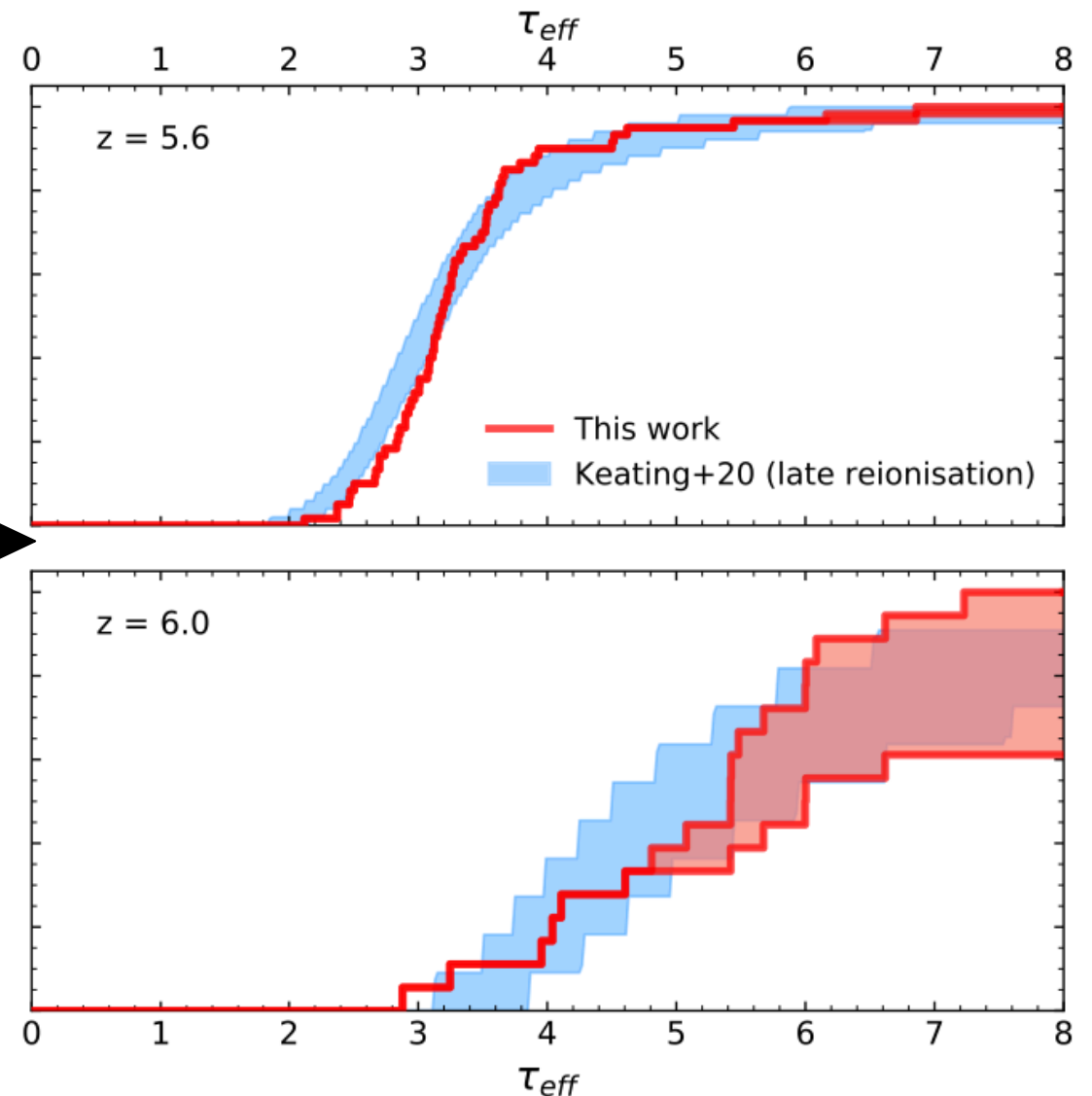
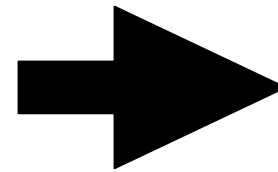
Bosman et al. (2022)

see also Becker+15, Bosman+18, Eilers+18

I. A late end to reionisation at $z \sim 5.3$



Credit: E. Puchwein/Sherwood collaboration



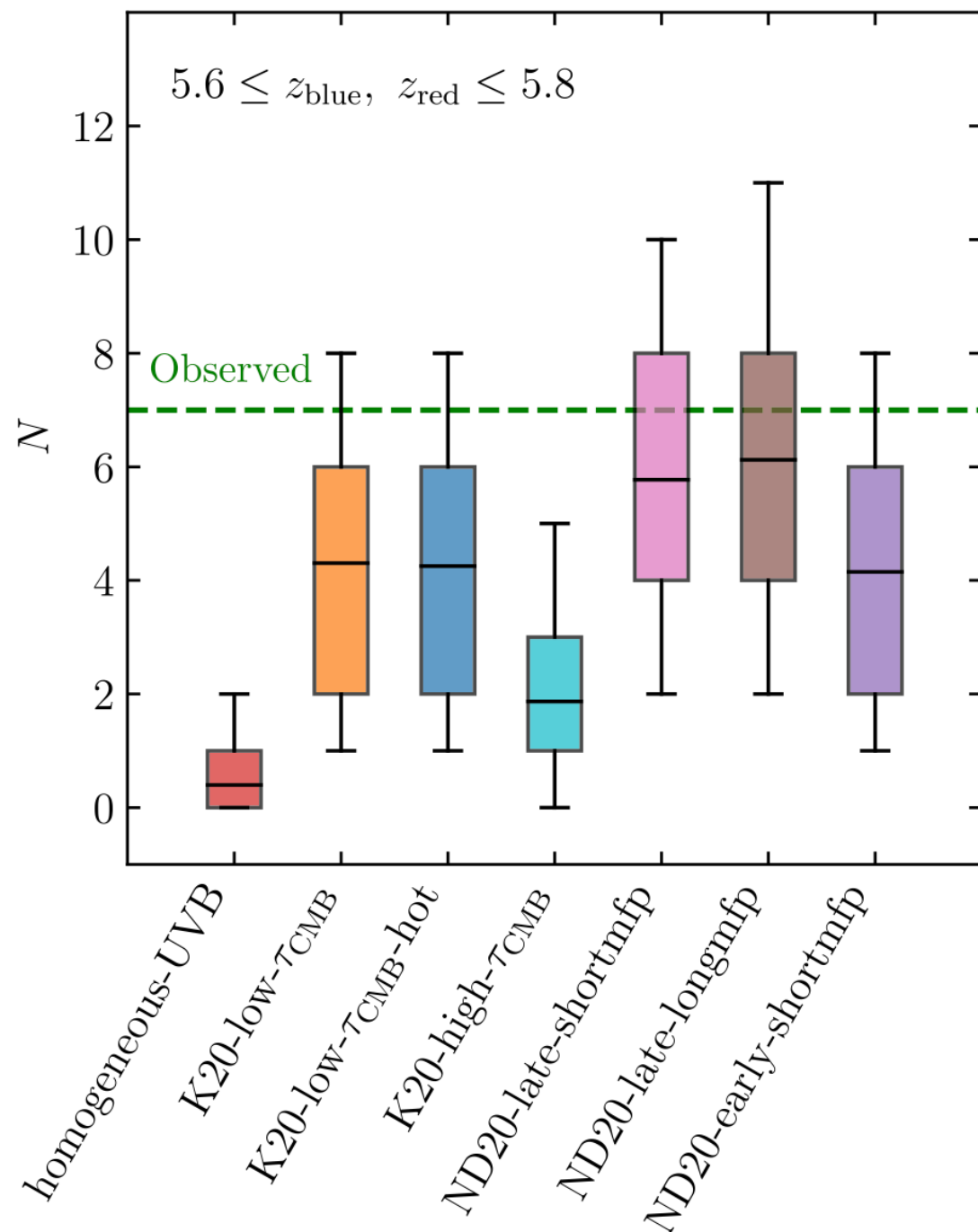
Bosman et al. (2022)

see also Becker+15, Eilers+18, Kulkarni+19, Keating+20, Nasir & D'Aloisio 20, Garaldi+22, Asthana+24, although see also Meiksin & Puchwein 24

I. Number of Ly- β dark gaps



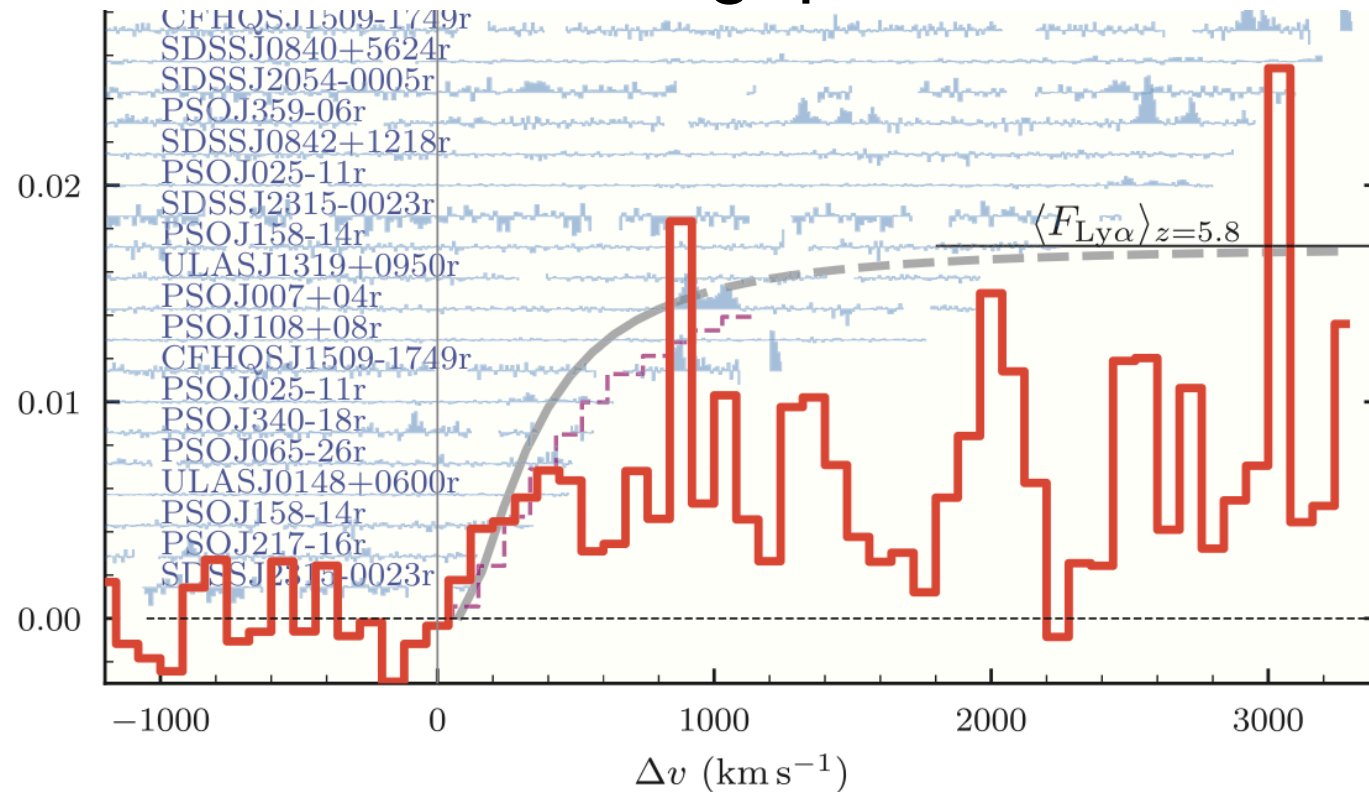
- Number and redshift distribution of Ly- β forest dark gaps requires late-end reionisation or large fluctuations in UV background (although see also [Gnedin+22](#))



Zhu et al. (2022),
see also Keating+20, Nasir & D'Aloisio 20

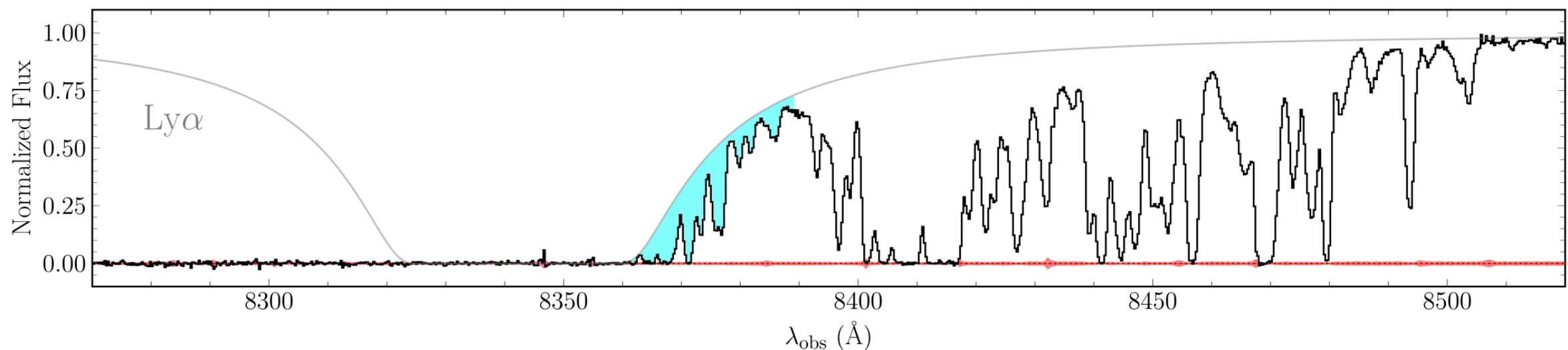
I. Ly- α damping wings at $z \sim 5.8$

Stack of 24 dark gaps



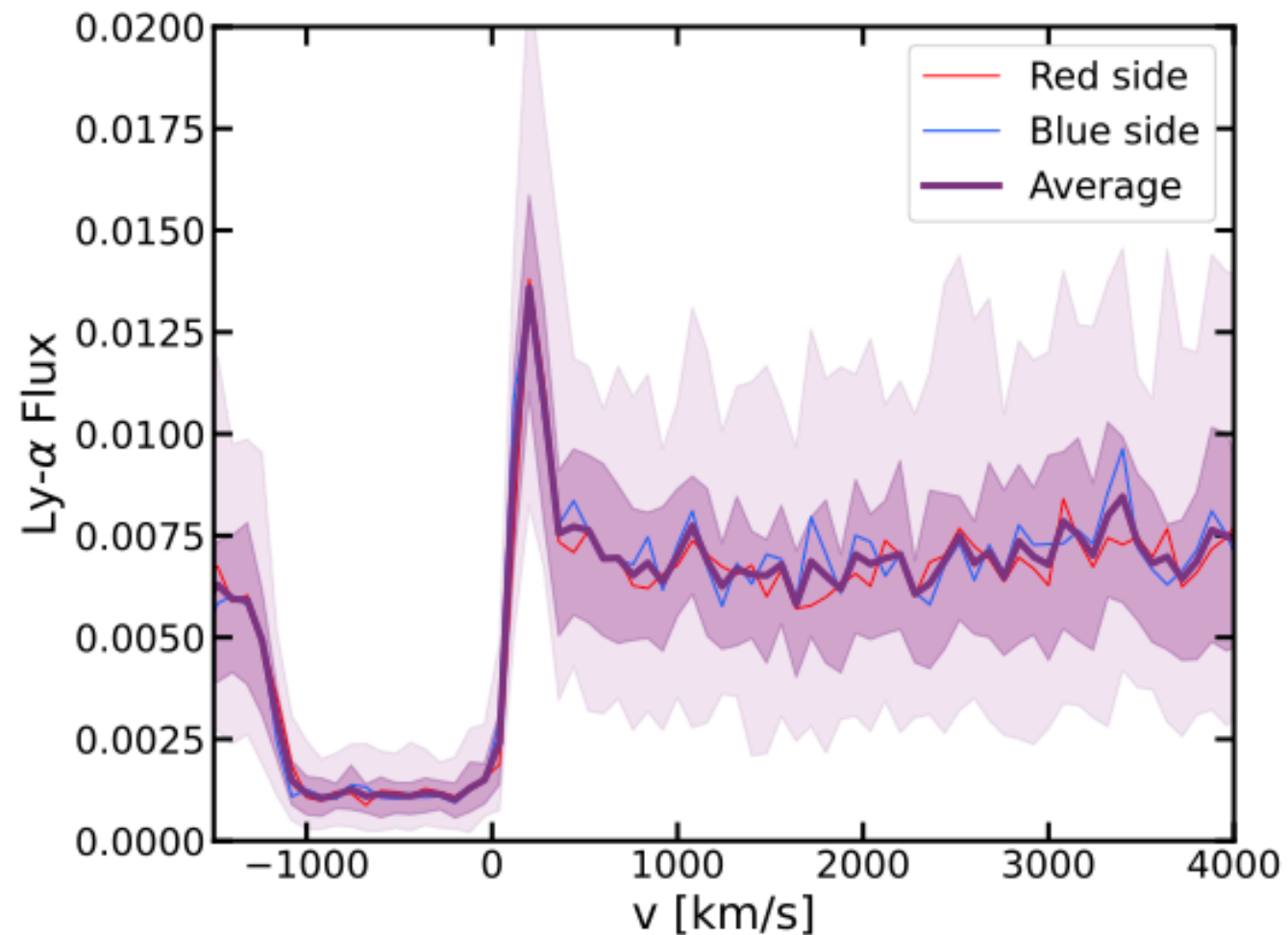
Zhu et al. (2024)
see also Spina+24

ULAS J0148+0600

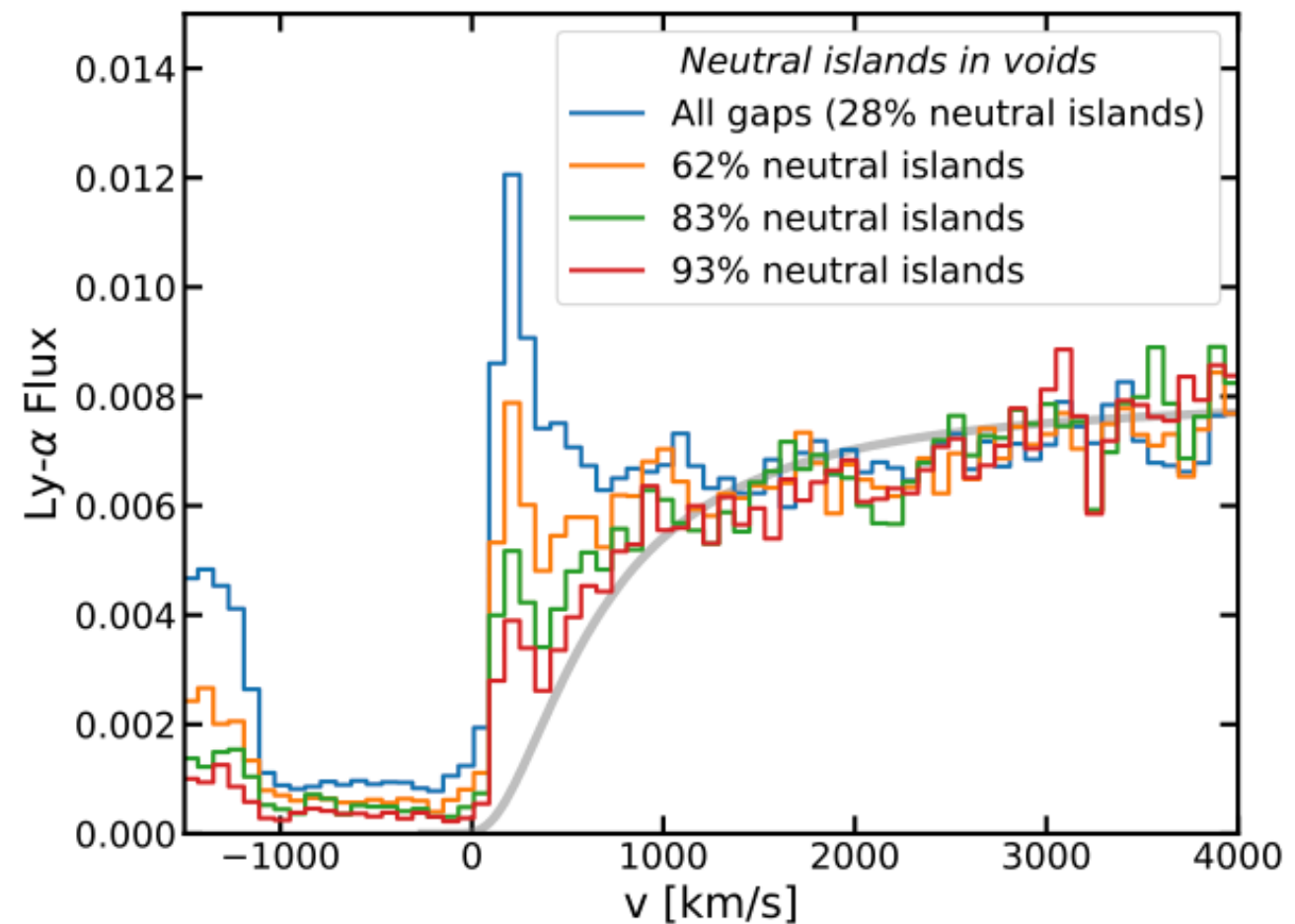


Becker et al. (2024)

I. No wings without neutral islands

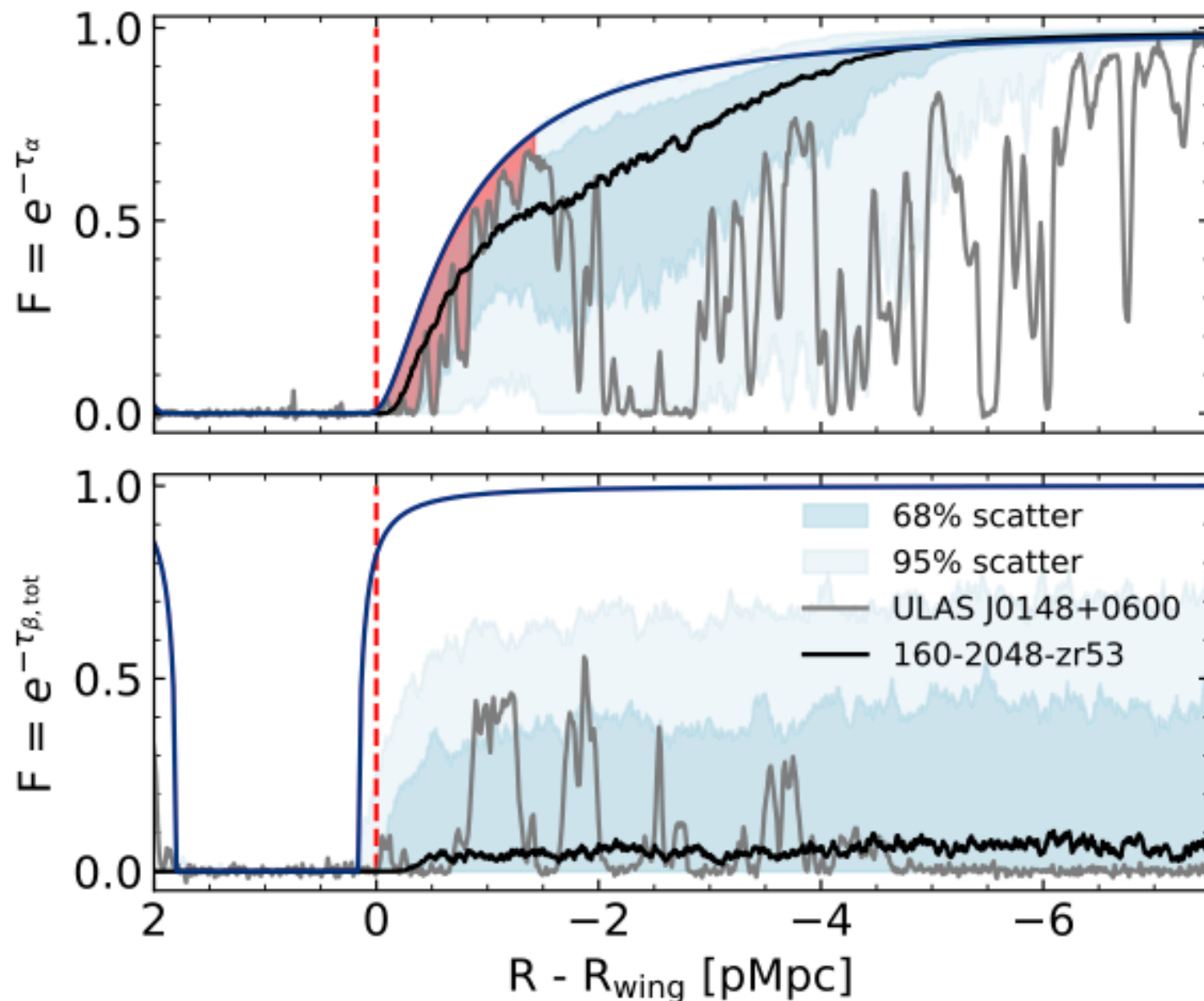


CROC simulation at $z=5.9$
 $\langle x_{\text{HI}} \rangle = 1.5 \times 10^{-4}$



Gnedin & Zhu (2025),
see also Malloy & Lidz (2015)

I. Physical model for the J0148 wing

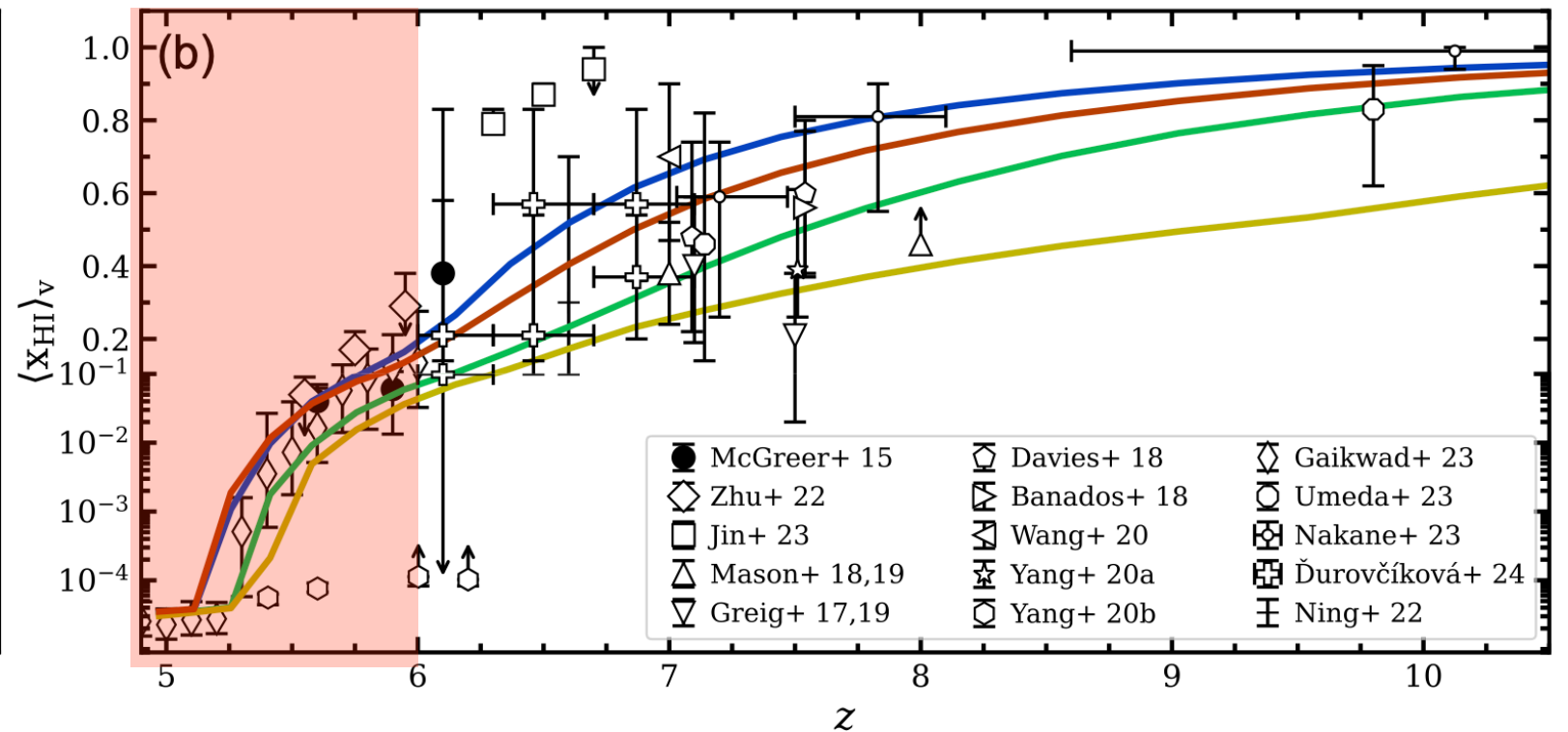
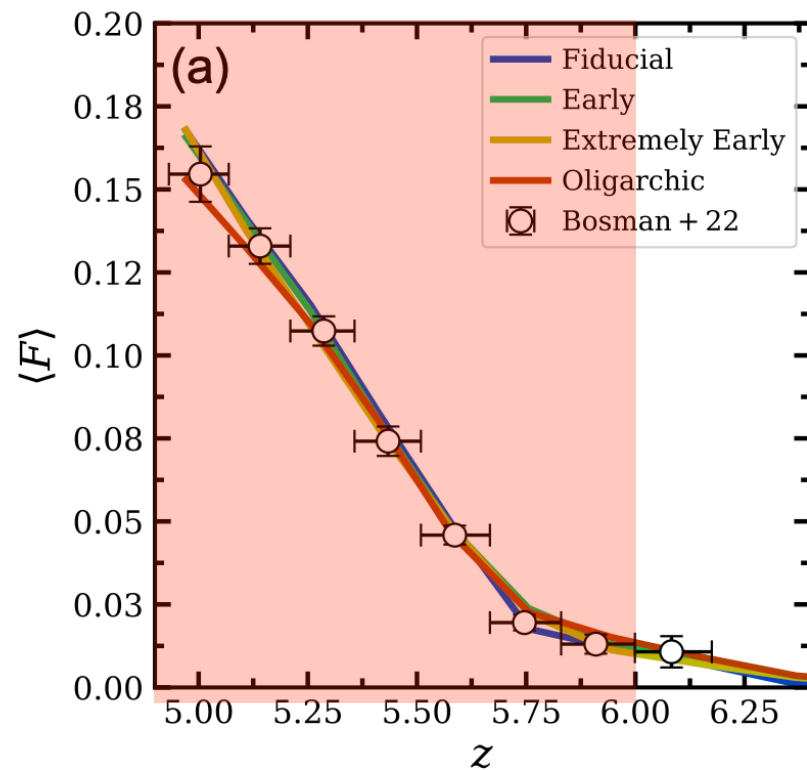
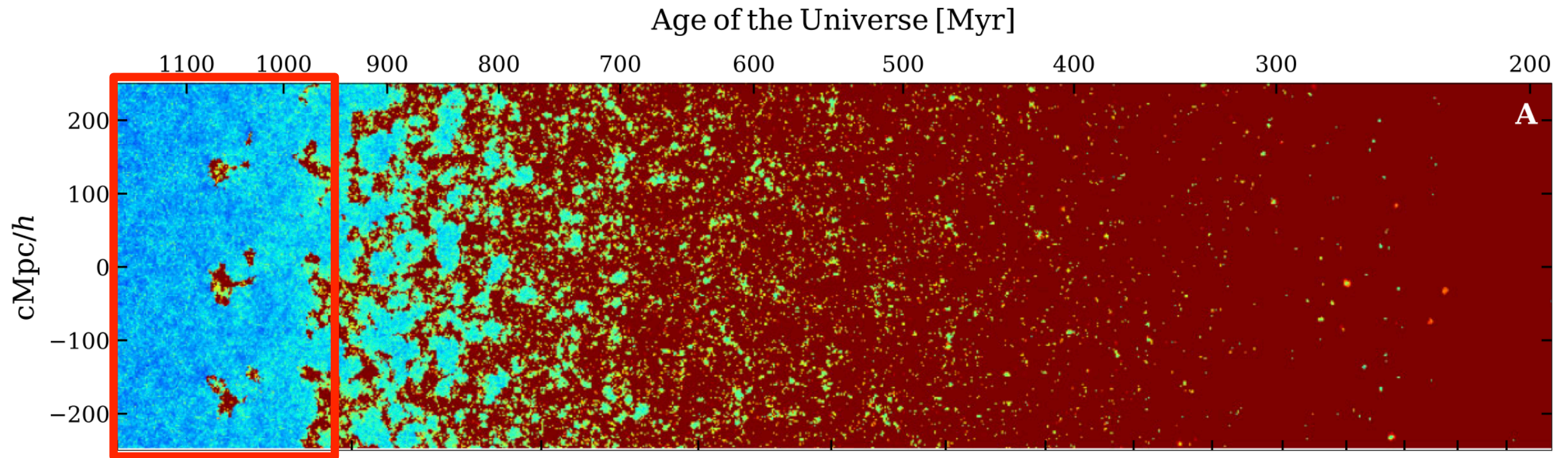


Sherwood-relics + quasar
RT model

- Require late-ending reionisation ($x_{\text{H1}}=0.14$ at $z=6$) for a chance ($p \sim 10^{-2}$) of matching observed profile.
- Requires neutral island close to blue edge of wing with transmission from ionised IGM photo-heated to $T \sim 40,000$ K by the quasar.

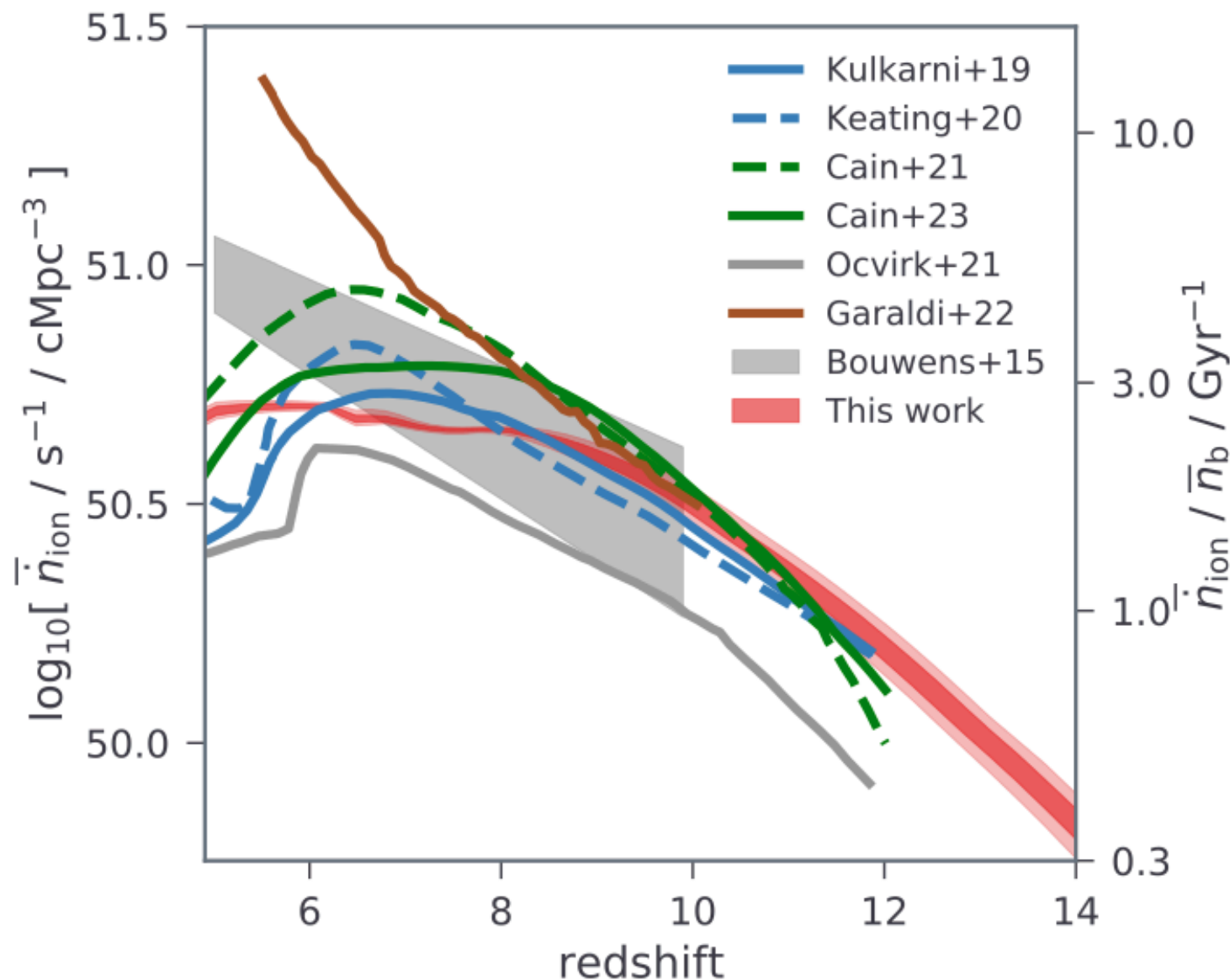
[Sawyer et al. \(2025\)](#)
[see also Becker+24](#)

I. A late end to reionisation



Asthana et al. (2024), see also Garaldi+22, Lewis+22, Qin+25

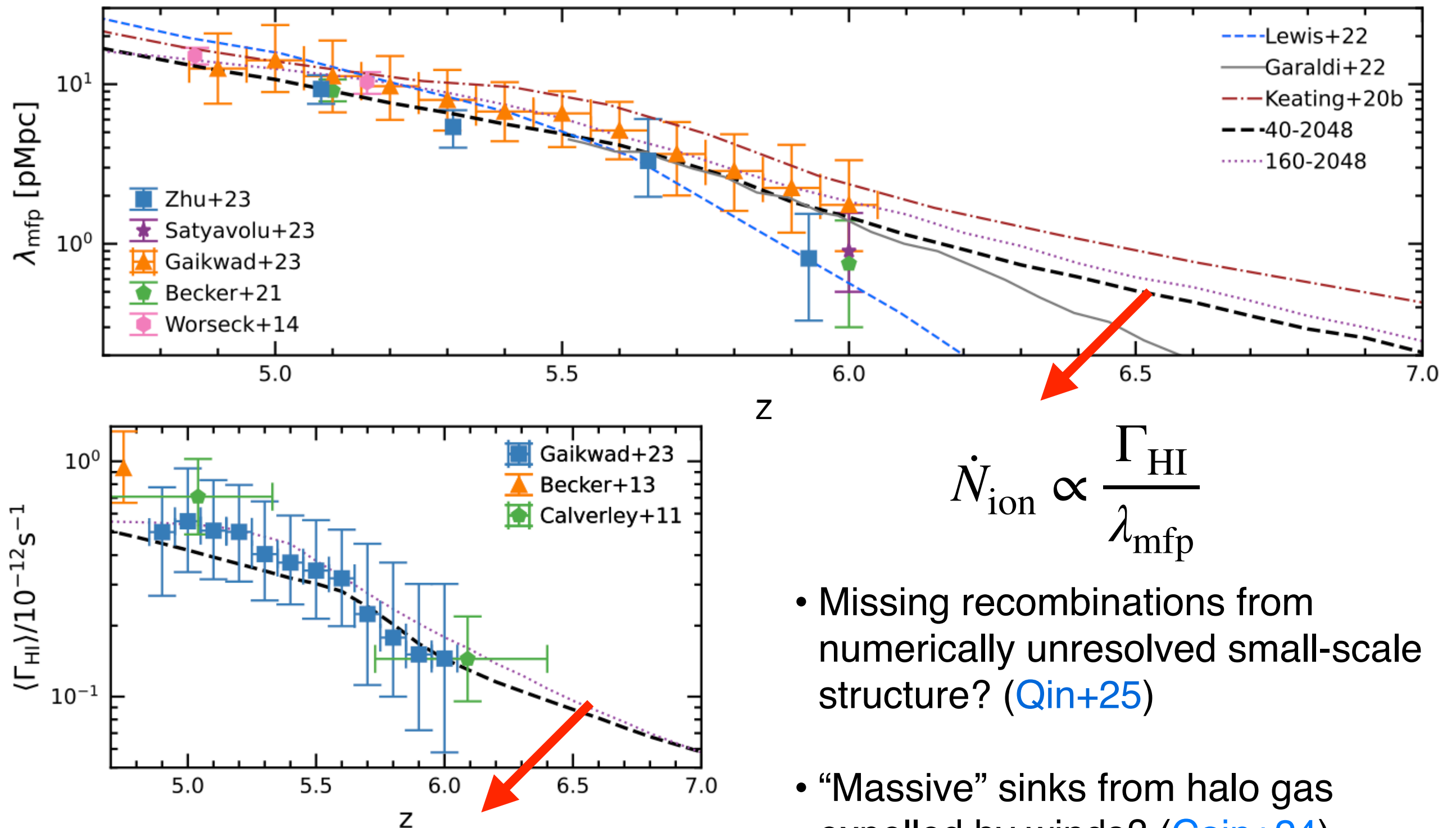
II. “Rise and fall” photon production



- Ly- α forest wants ~ 1 to 3 photons per H atom by $z=6$. “Photon-starved” end to reionisation / “soft-landing”
- Radiative transfer simulations struggle to avoid a “rise and fall” in $\dot{N}_{\text{ion}}$
- Physics (radiative feedback, [Ocvirk+21](#)) or numerics (unresolved sub-grid structure [Qin+25](#))?

[Qin et al. \(2025\)](#), see also [Cain+24](#)

II. “Rise and fall” photon production



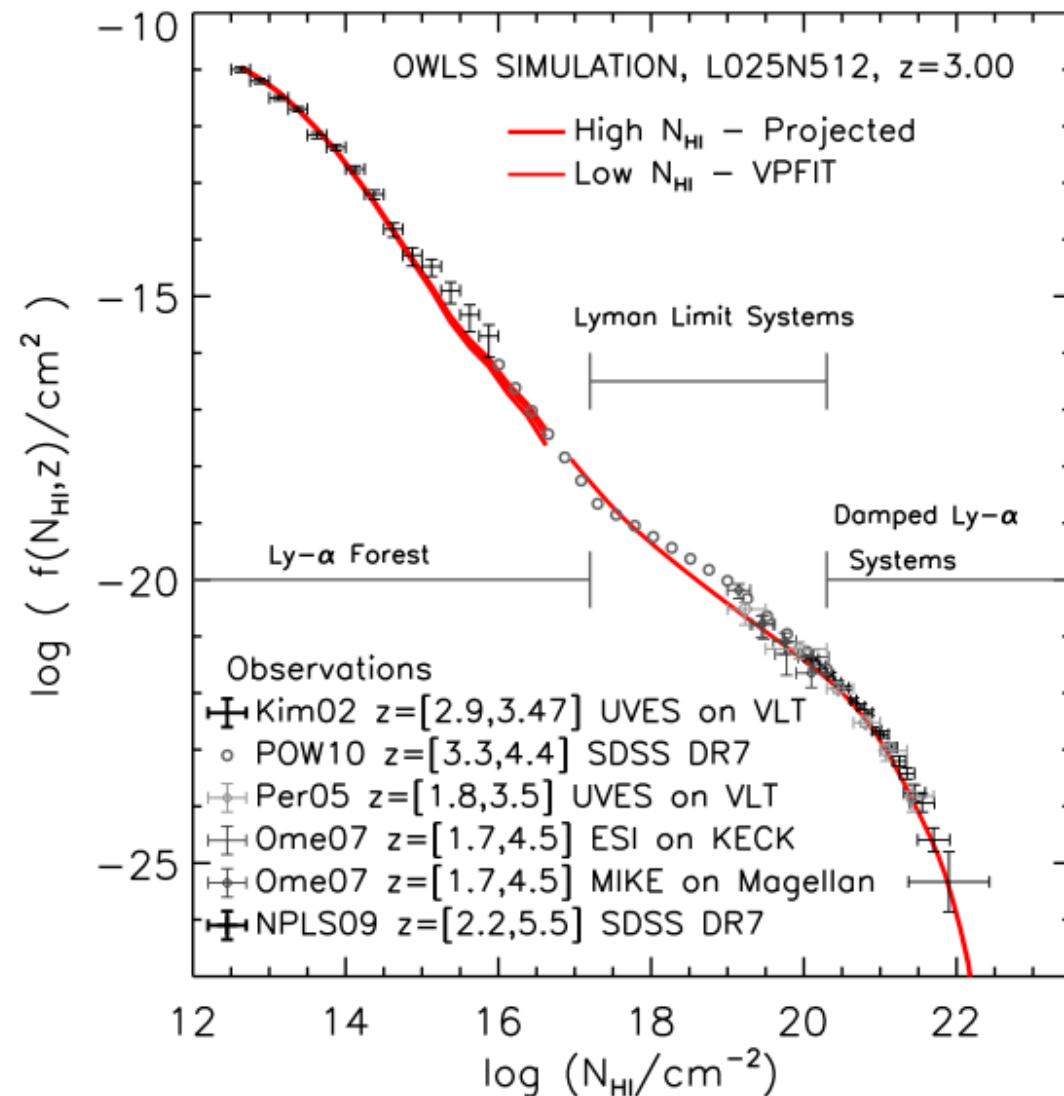
$$\dot{N}_{\text{ion}} \propto \frac{\Gamma_{\text{HI}}}{\lambda_{\text{mfp}}}$$

- Missing recombinations from numerically unresolved small-scale structure? (Qin+25)
- “Massive” sinks from halo gas expelled by winds? (Cain+24)

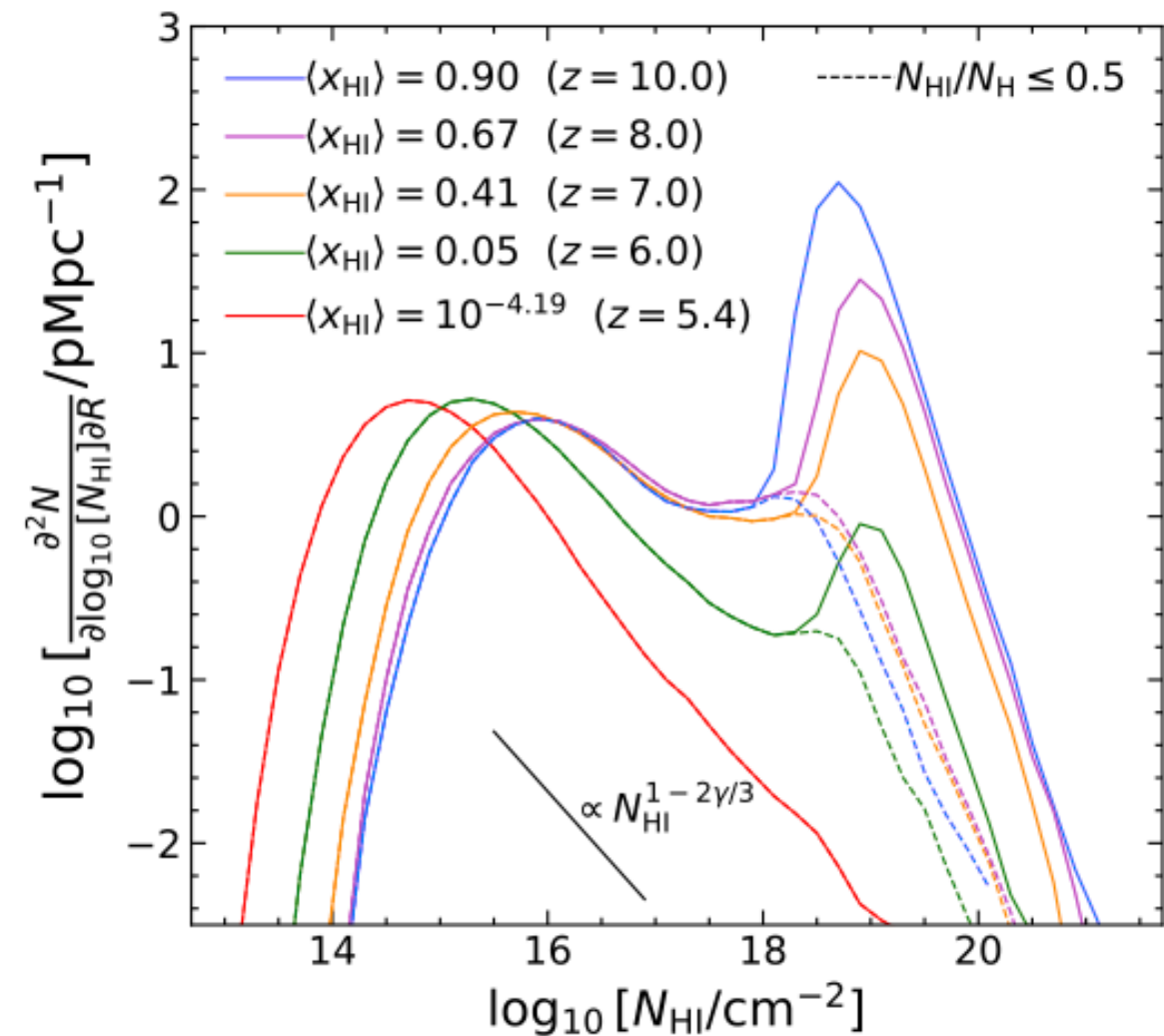
Feron et al. (2024)

II. The column density distribution

HI column density distribution



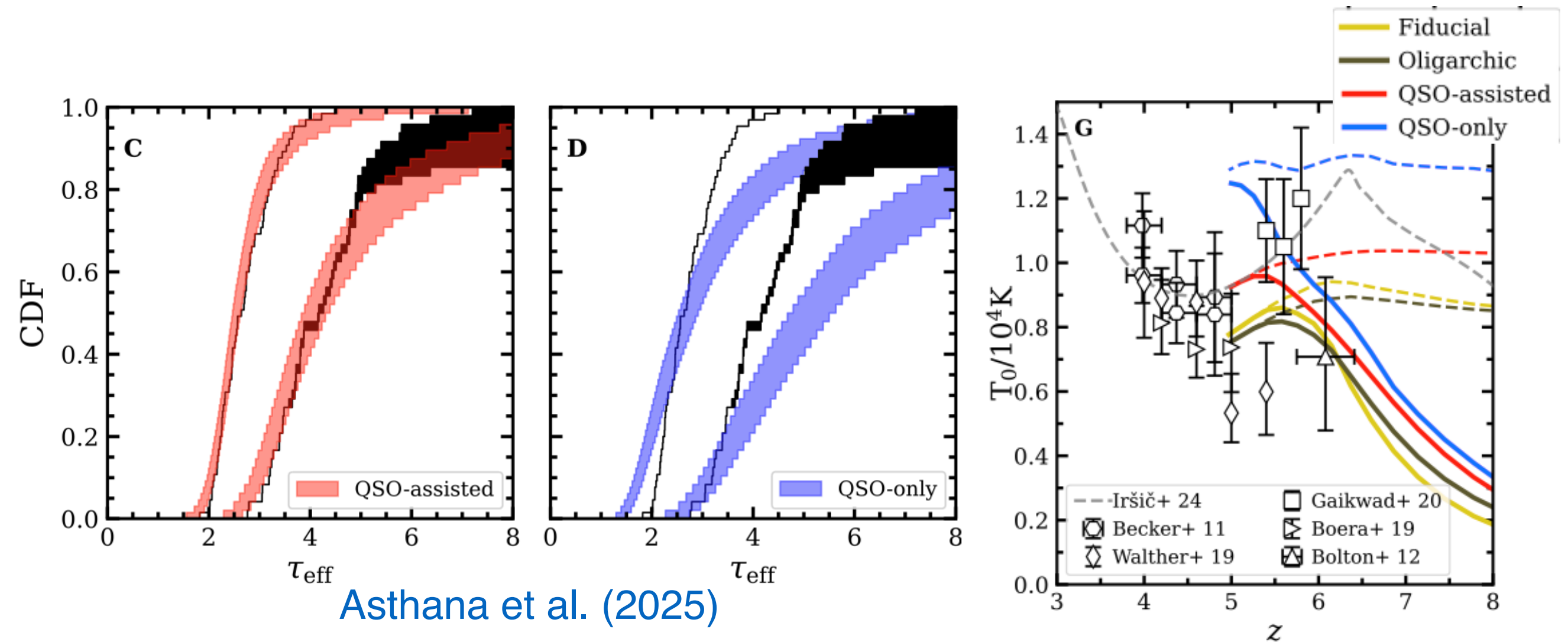
Altay et al. (2011)



Feron et al. (2024)

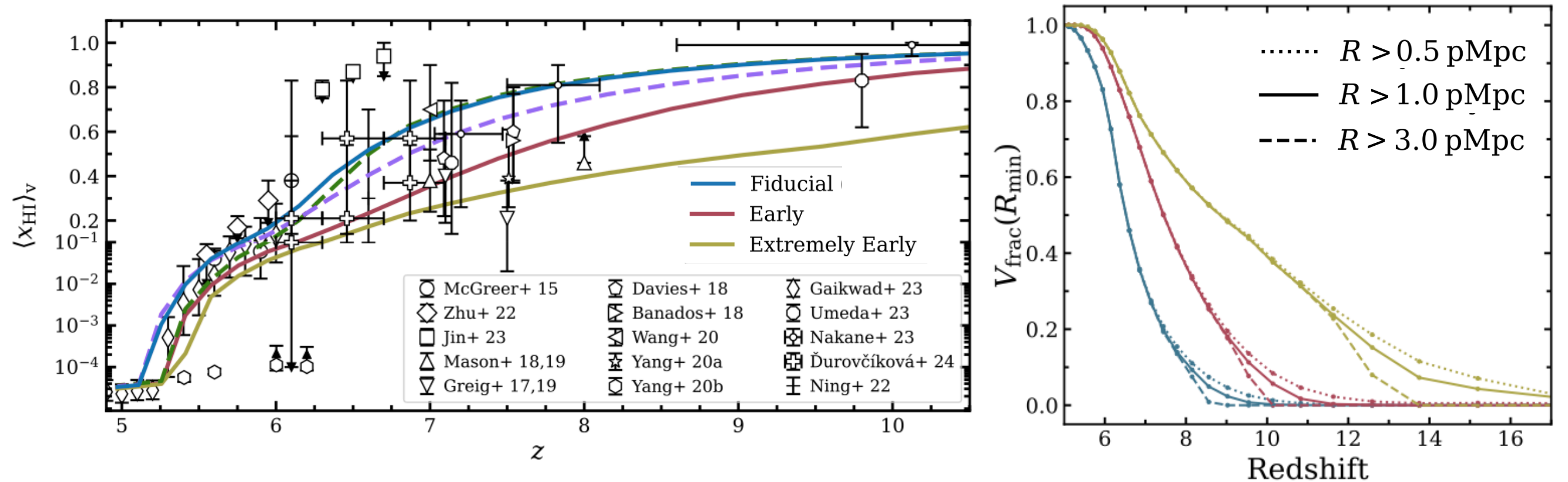
- By $z=6$ most of the Lyman-limit opacity is associated with the (numerically fairly well-resolved) photo-ionised gas (see also [Rahmati & Schaye 2018](#))

II. The AGN contribution



- What about faint AGN? ([Harikane+23](#), [Grazian+24](#), [Maiolino+24](#))
- QSO-only model fails to match τ_{eff} distribution (but QSO assisted = 17% of total emissivity from AGN, works well).
- Hard to drive reionisation with AGN and match temperature unless He-II ionising photons do not escape (cf. [Madau+24](#))

II. LAEs and early start + late end



Asthana et al. (2024)

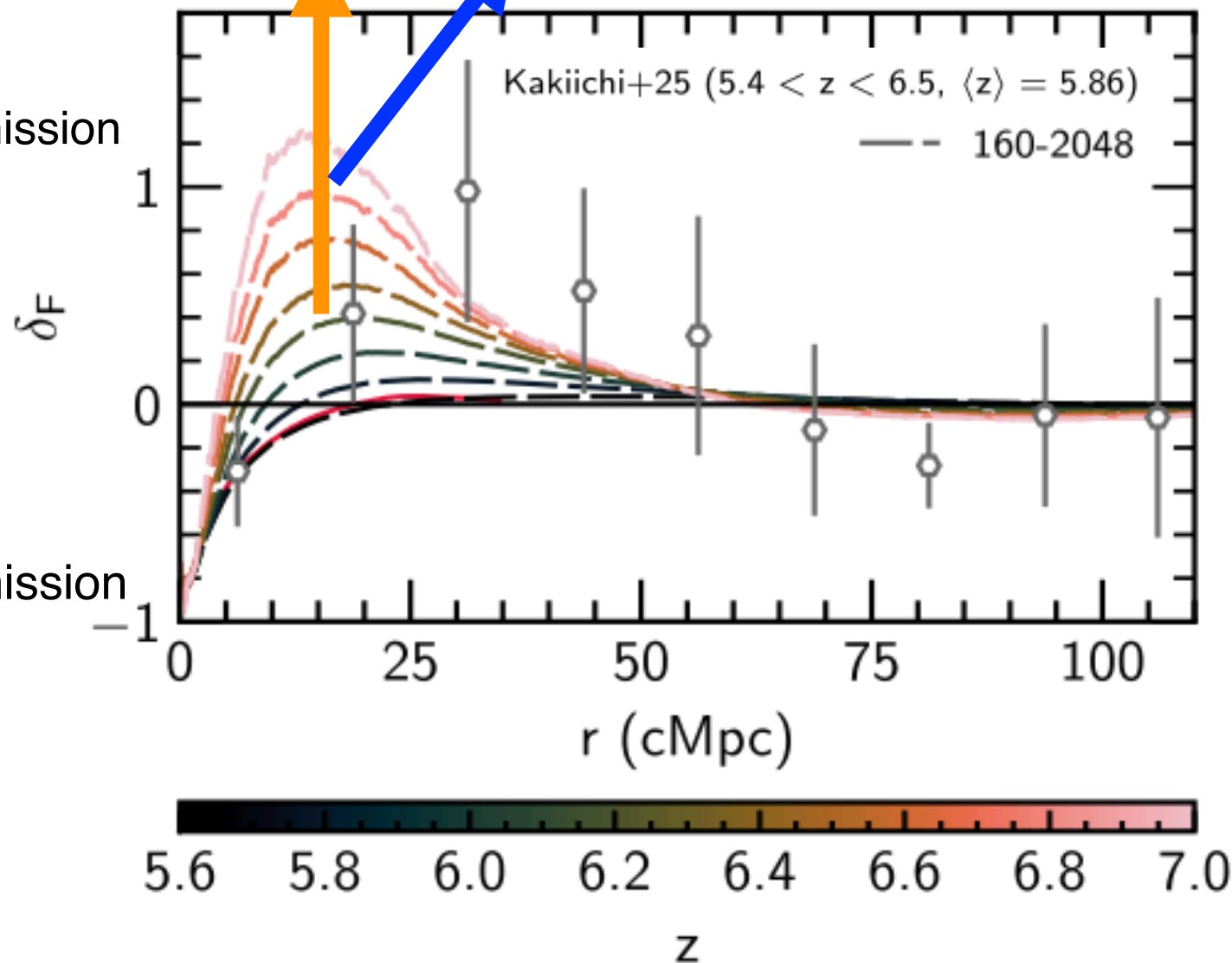
- Models all match Ly- α τ_{eff} distribution by design.
- For fiducial model, at $z \sim 6.5$ ($z=11$) for $x_{\text{H I}} \sim 0.5$ (0.95), 50% ($\sim 0\%$) of IGM filled with bubbles > 1 pMpc (see also [Lu+24](#), [Qin & Wyithe 2025](#)).
- Reconciling with $z > 10$ LAEs (e.g. [Bunker+23](#), [Witstok+24](#)) may require lower $x_{\text{H I}}$ / earlier start, but can lead to tension with Planck τ_e (see also [Cain+25](#) for kSZ tension for larger τ_e).

II. The galaxy-Ly- α correlation

Neutral fraction Halo mass

More
transmission

Less
transmission

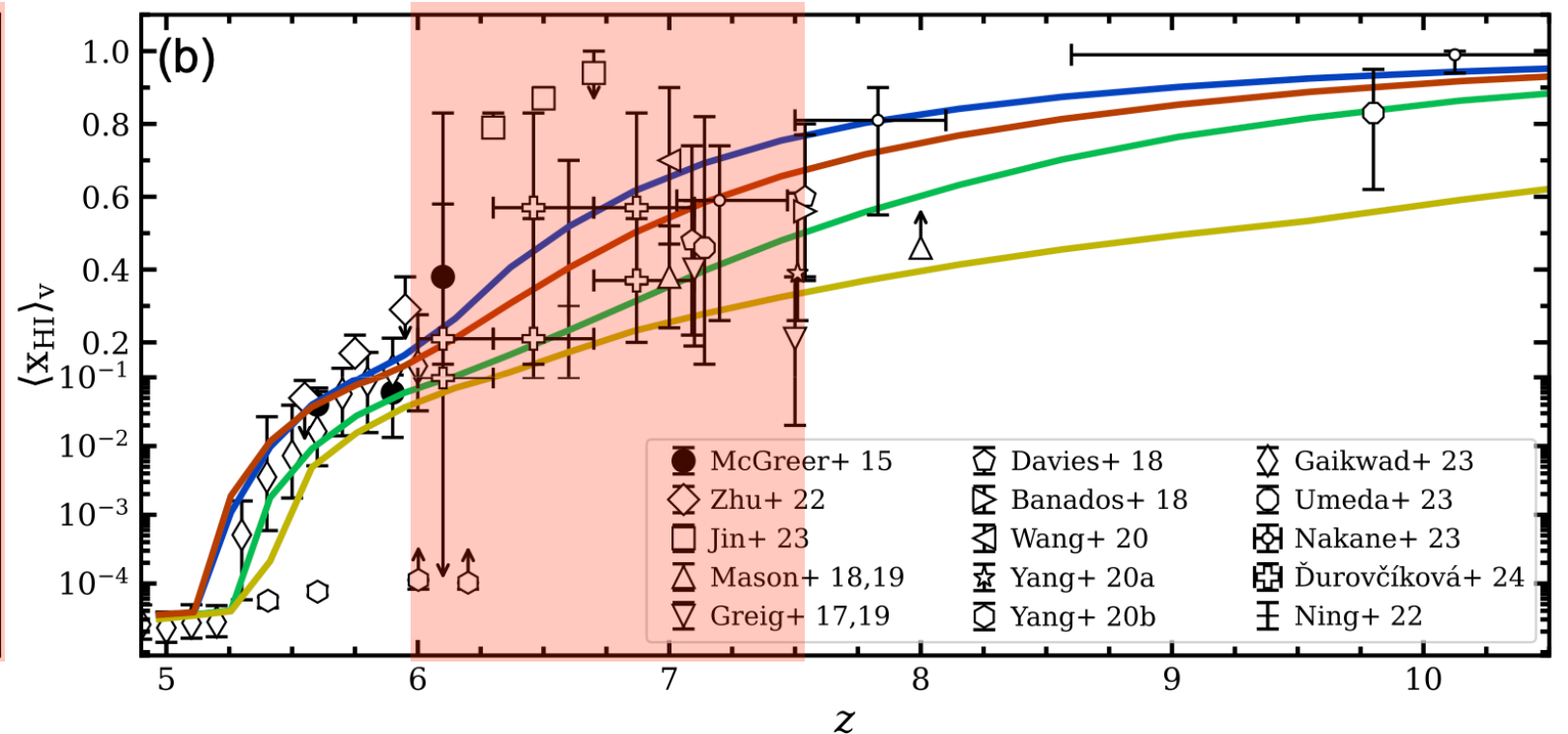
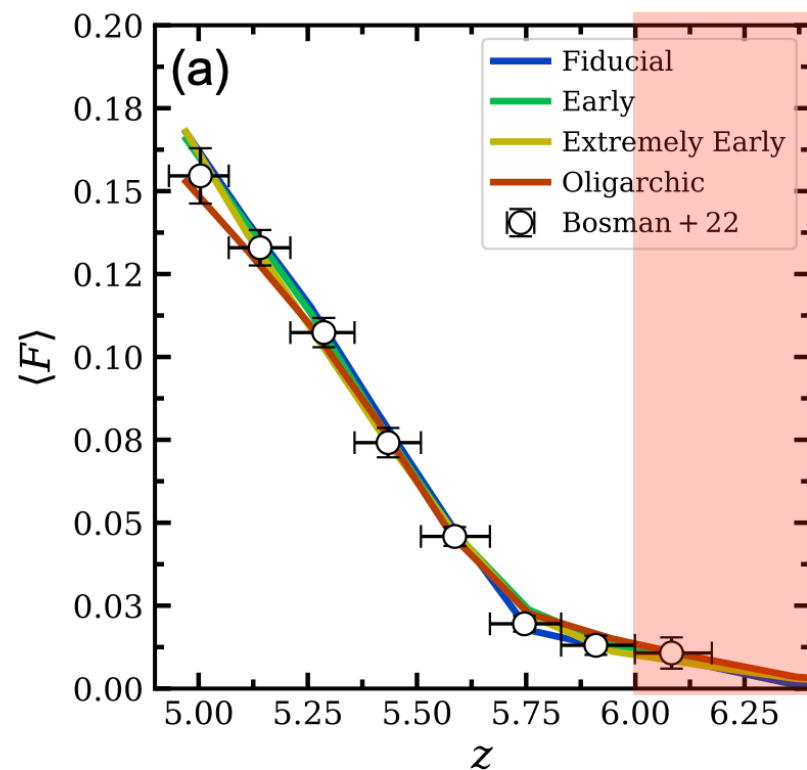
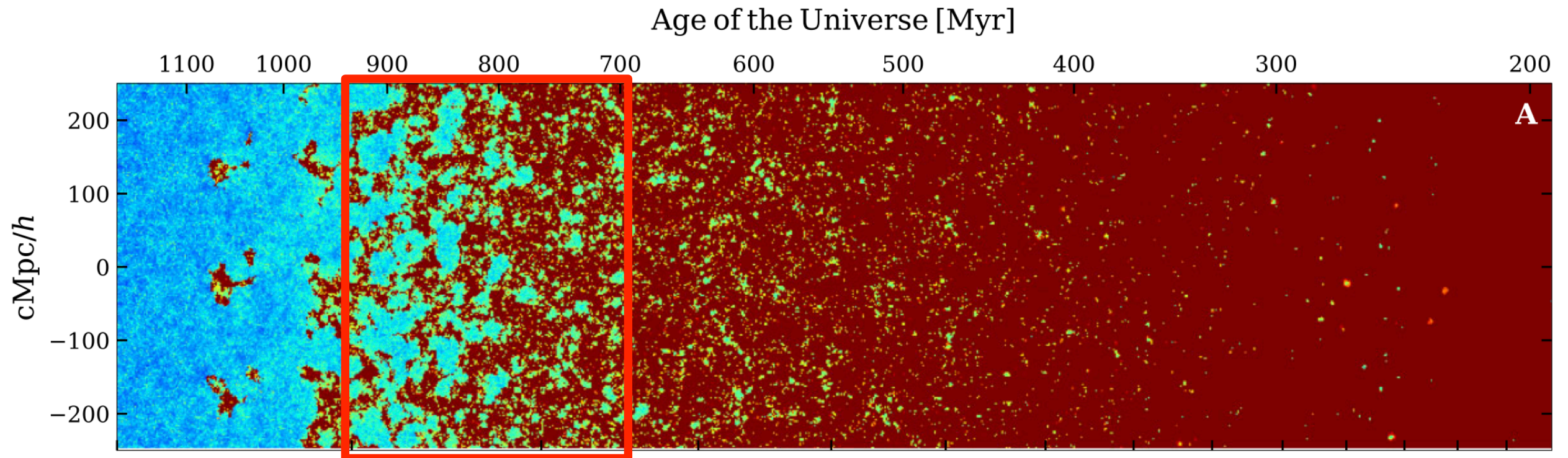


- Reasonable ($\sim 1.5\sigma$) agreement with ASPIRE (Kakiichi+25) for haloes $> 10^{10} M_\odot/h$
- But, peak offset by ~ 20 cMpc.
- See talk by Luke Conaboy

Conaboy et al. (2025)

see also Garaldi & Bellscheidt 24, Kashino+25 (EIGER)

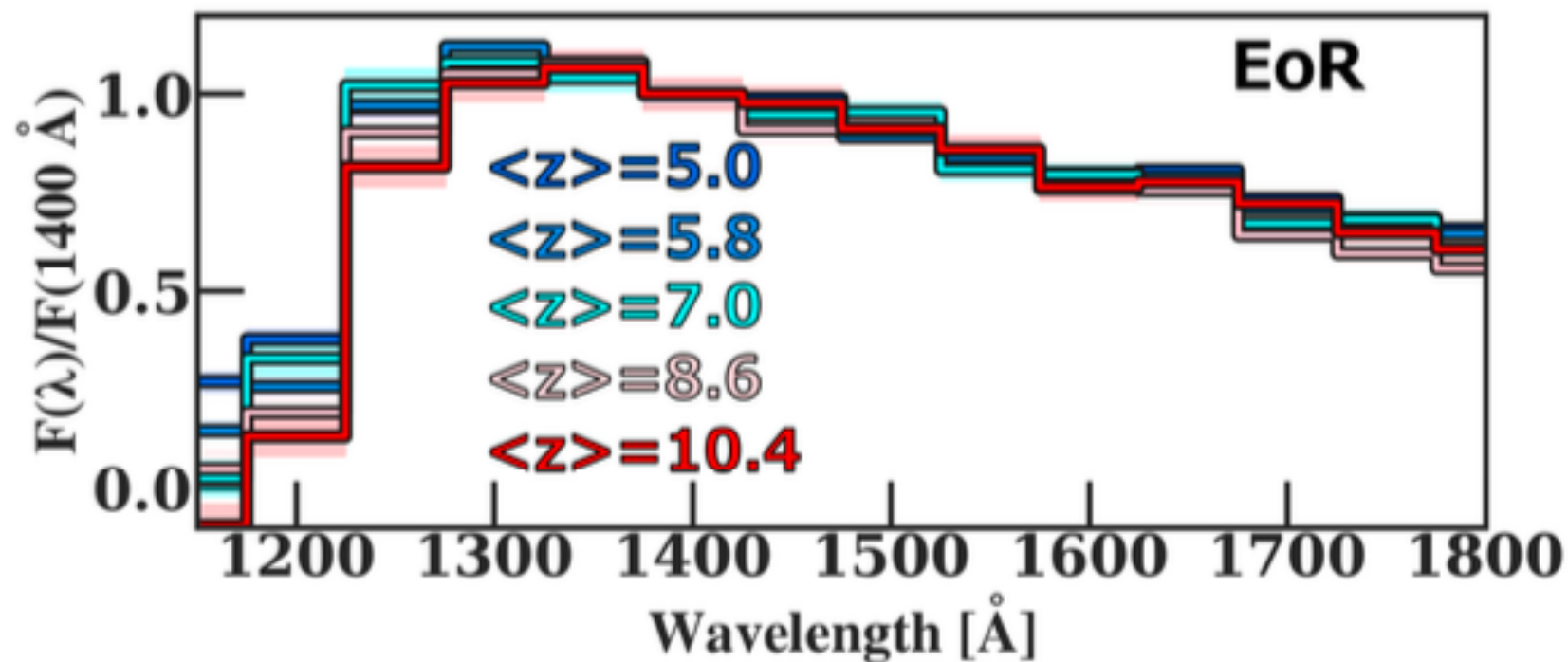
III. Ly- α absorption at $z > 6$



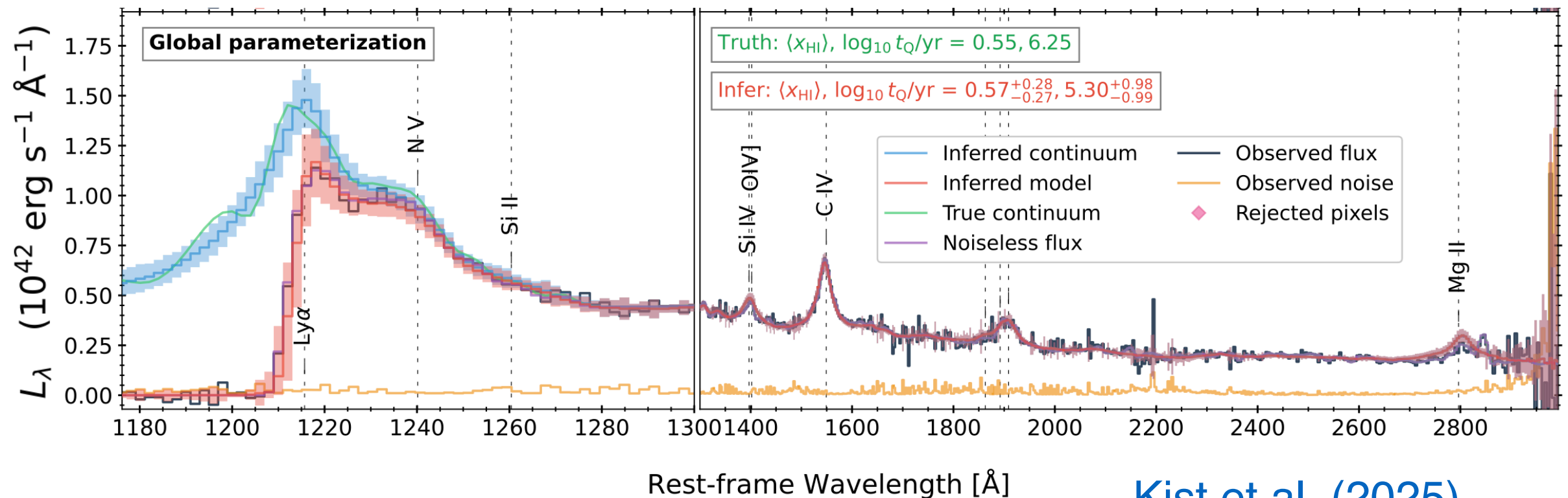
Asthana et al. (2024), see also Garaldi+22, Lewis+22, Qin+25

III. Damping wings (galaxies+QSOs)

7

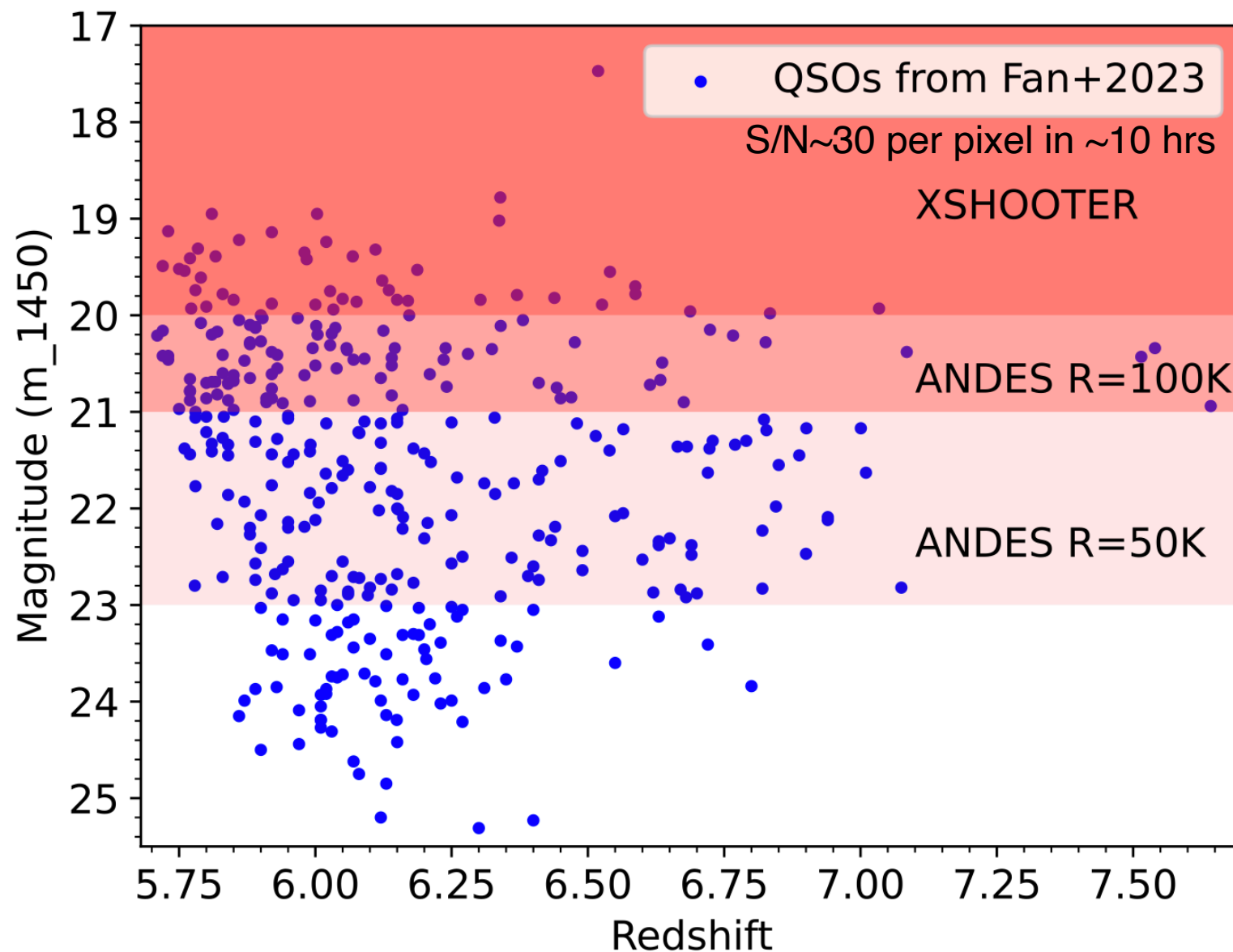


Umeda et al. (2025)



Kist et al. (2025)

III. ELT/ANDES



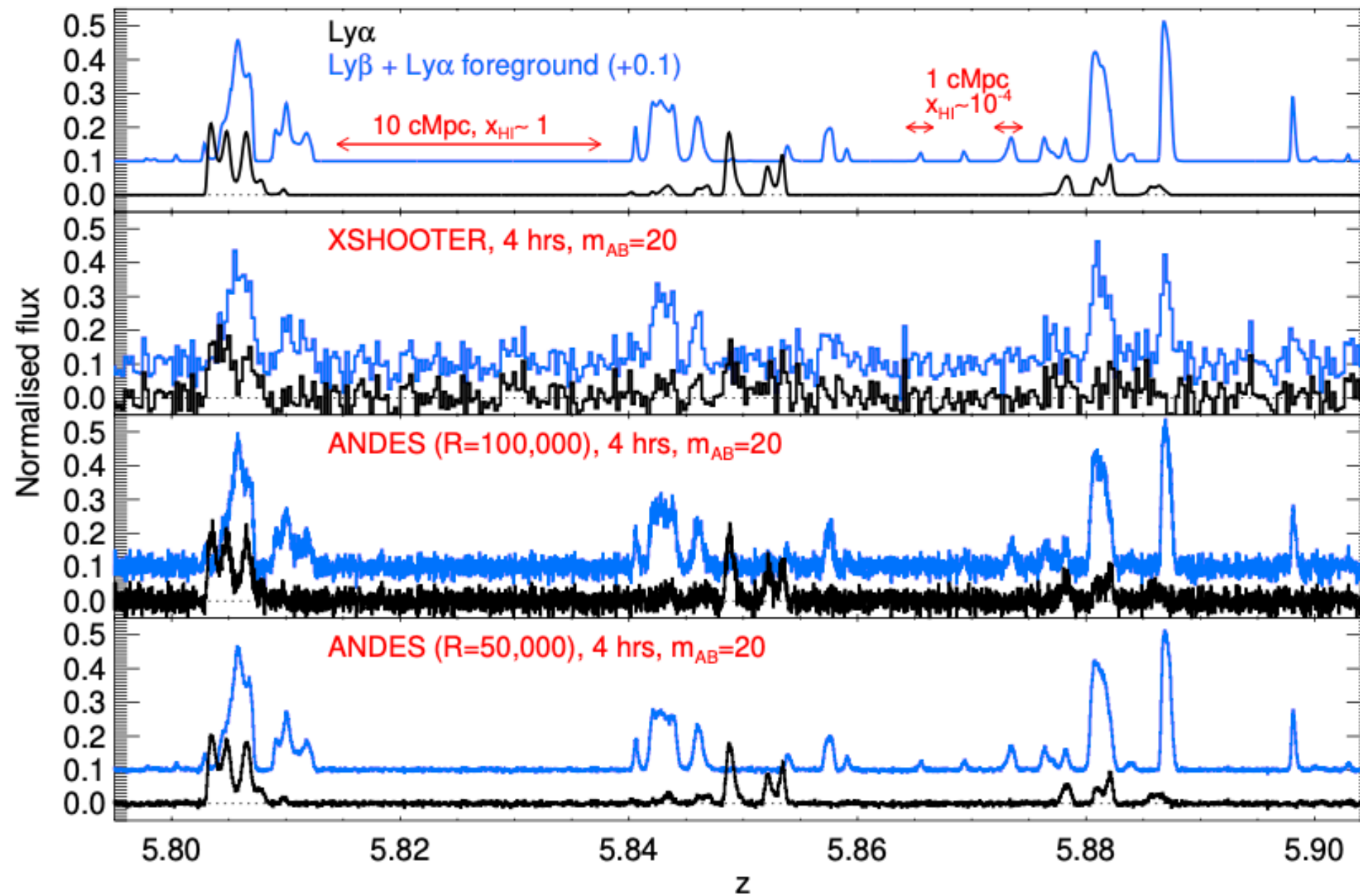
Key areas:

- QSO proximity/near-zones
- Lyman series transmission / patchiness of reionisation
- Metals, Pop-III signatures
- IGM-galaxy correlation
- Dark matter models (warm/cold)
- and more...

D'Odorico et al. (2024)

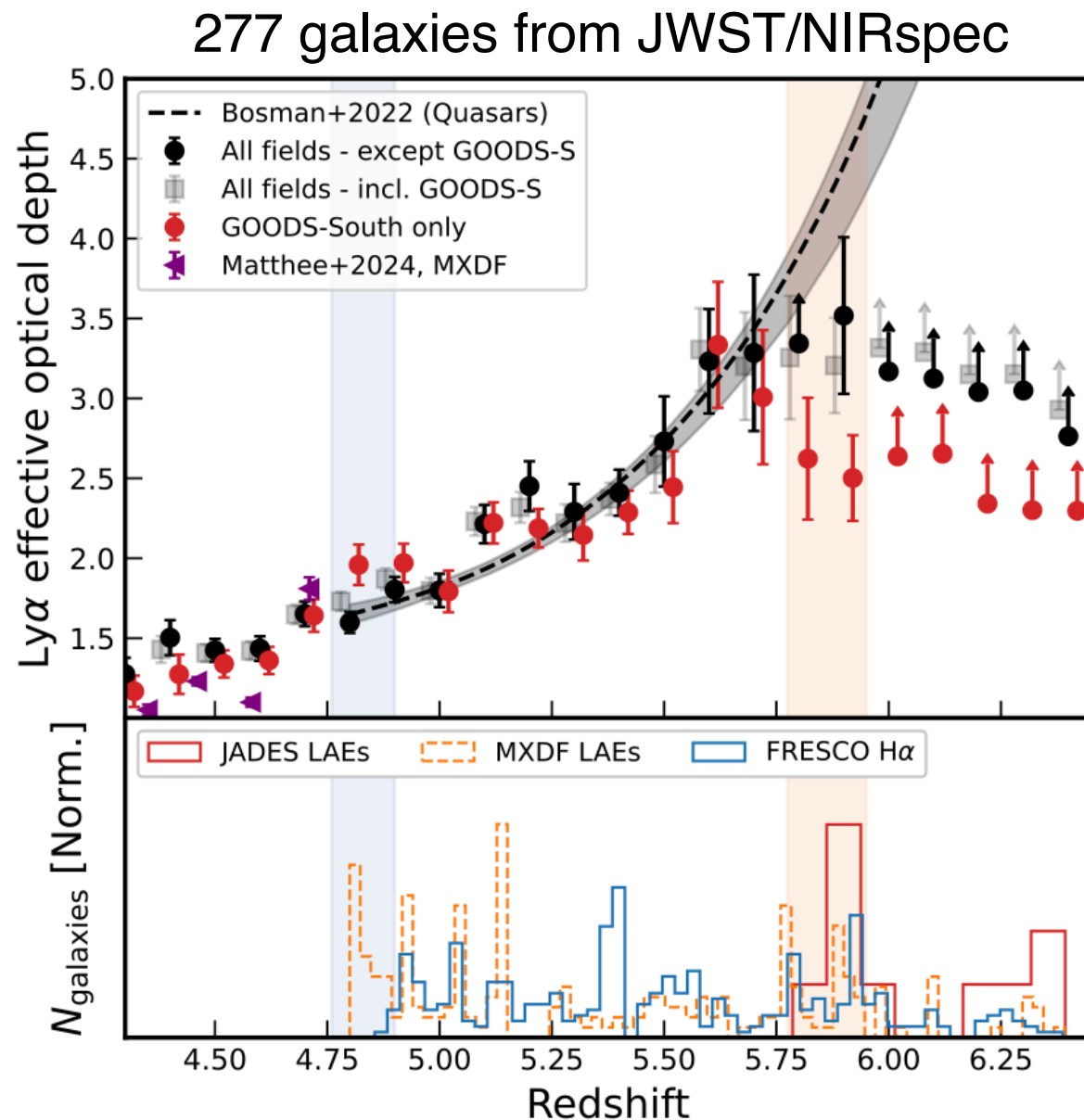
In the nearer term, also GHOSTLy (Farina, Bosman et al.) for $R > 50,000$ at $4 < z < 5$

III. $\text{Ly-}\alpha$ + $\text{Ly-}\beta$ with ELT / ANDES



D'Odorico et al. (2024)

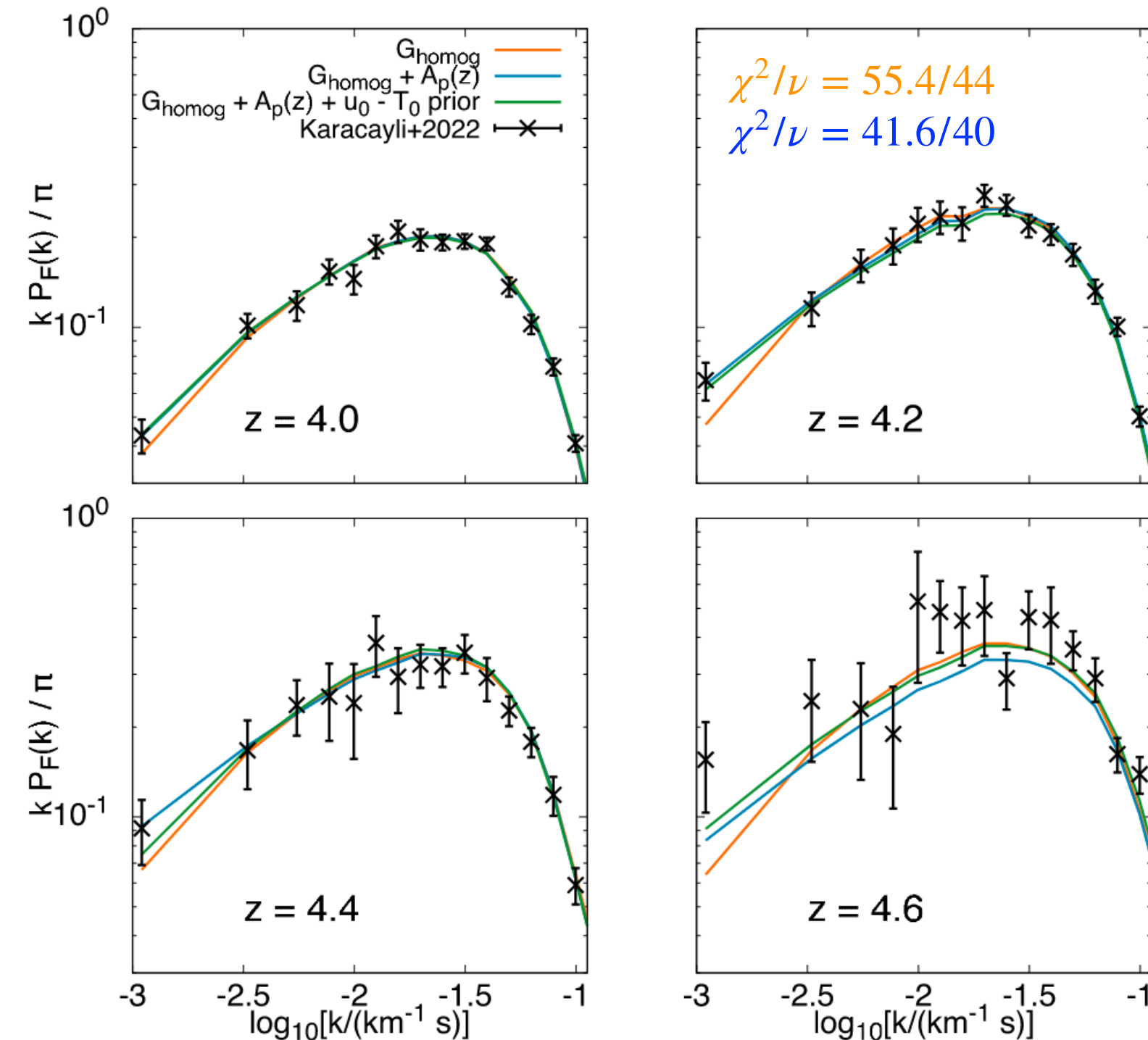
III. JWST/NIRspec IGM spectroscopy



- Advantage: galaxies are orders of magnitude more numerous than bright QSOs. Useful for correlating galaxies with IGM absorption.
- Challenge: they are faint, with low resolution spectra.

Meyer et al. (2025)
see also Giri+25

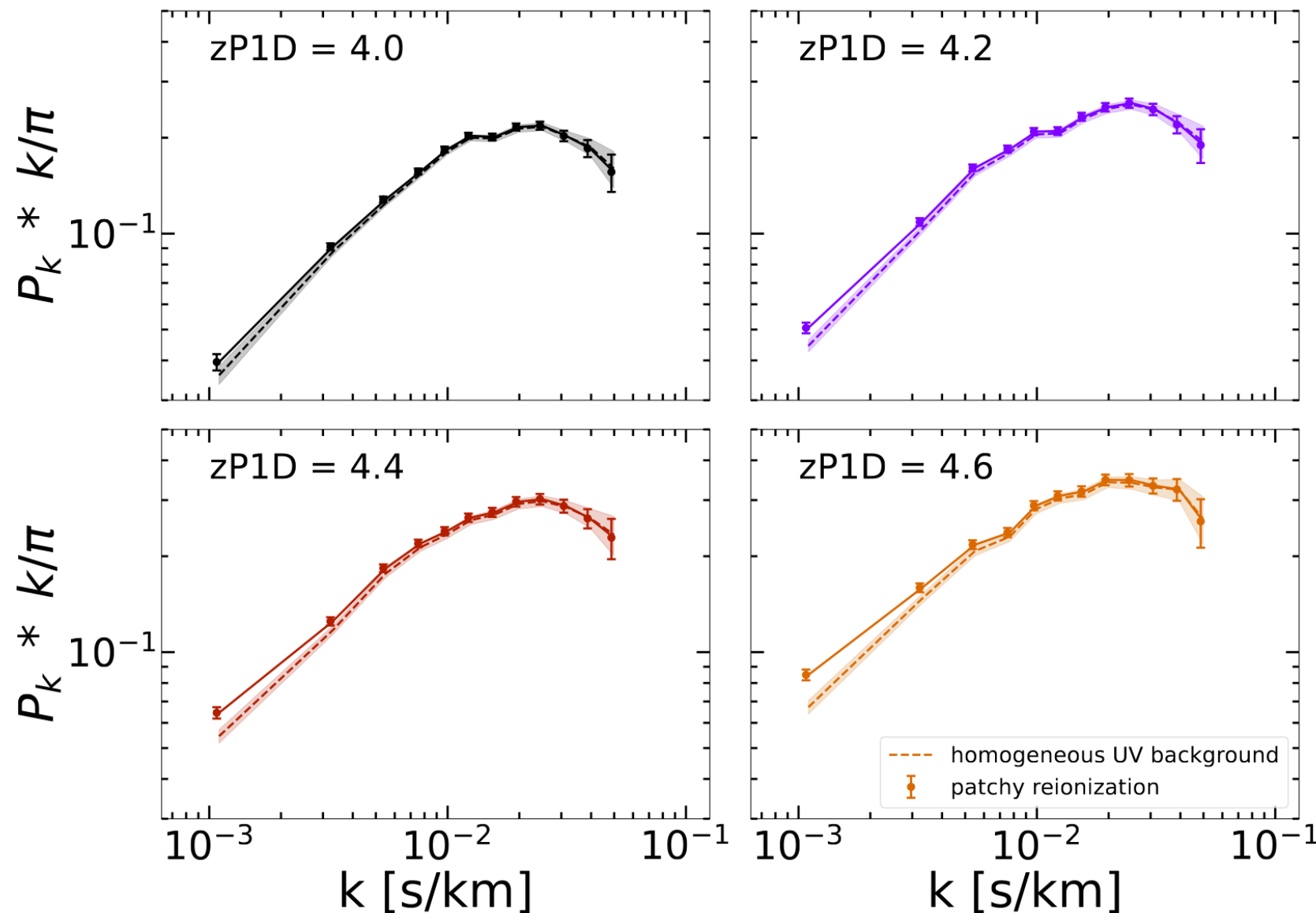
III. Post-reionisation $P(k)$



- 2.7σ preference for an enhancement in power at large scales relative to uniform UVB;
- Possible signature of relic post-reionisation temperature fluctuations? (see also [D'Aloisio+18](#) for an earlier $\sim 2\sigma$ hint at $z=5.4$).

Molaro et al. (2023), see also Hirata+18, Montero-Camacho+24, Cain+24 for hydrodynamics driven temperature fluctuations

III. Massive spectroscopic surveys



Ma et al. (in prep.)

- Errors (stat. + sys.) $< 5\%$ for 10,000 sq. deg. QSO survey at $R \sim 5000$
- Joint analysis with flux distribution should further sharpen constraints on the timing of the end stages of reionisation.
- Massive spectroscopic surveys can test this:
DESI, WEAVE-QSO, 4MOST/ByCycle, WST...

Ly- α absorption: a summary

- **Progress:** Growing agreement that reionisation ends at $z < 6$ (τ_{eff} distribution, dark gaps, damping wings at $z \sim 5.8$ from XQR-30, modelling)
- **Challenges:** photon sinks and dynamic range, AGN contribution and IGM temperature, LAEs and reionisation duration, galaxy-IGM correlation.
- **Looking ahead:** Higher resolution and fainter quasar targets with ELT/ANDES (+more targets from Euclid). Nearer term, GhostLY ($R \sim 50,000$) and massive spectroscopic surveys ($R < 5000$). Ongoing JWST/NIRspec galaxy spectroscopy.